# FTSA
# TESLA
# 2026

Esin FİRUZAN

# TESLA

Bu sunumdaki tüm çözümler 2020-2024 verisi içindir.

```
> df_tesla <- getSymbols('TSLA',src='yahoo',auto.assign=FALSE,from = "2020-06-01", to = "2024-11-05", inde
x.class = "POSIXct", period = "daily")
> head(df_tesla)
           TSLA.Open TSLA.High TSLA.Low TSLA.Close TSLA.Volume TSLA.Adjusted
2020-06-01  57.20000  59.93333 56.94000   59.87333   224092500      59.87333
2020-06-02  59.64667  60.57733 58.06667   58.77067   203484000      58.77067
2020-06-03  59.20800  59.86267 58.67333   58.86400   119242500      58.86400
2020-06-04  59.32533  59.71667 57.22933   57.62533   133315500      57.62533
2020-06-05  58.52267  59.10133 57.74667   59.04400   117178500      59.04400
2020-06-08  61.26667  63.33333 60.61067   63.32800   212620500      63.32800

> class(df_tesla)
[1] "xts" "zoo"
> nrow(df_tesla)
[1] 1116
> tail(df_tesla,2)
           TSLA.Open TSLA.High TSLA.Low TSLA.Close TSLA.Volume TSLA.Adjusted
2024-11-01    252.04     254.0   246.63     248.98    57544800        248.98
2024-11-04    244.56     248.9   238.88     242.84    68662200        242.84
```

# TESLA

```
plot(df_tesla$TSLA.Close,main = 'TESLA Hisse Fiyati',subset='2010::2024')
```



# TESLA


TESLA Hisse Fiyati
2020-06-01 / 2024-11-04
400
300
200
100
Jun 01 2020
Dec 01 2020
Jun 01 2021
Dec 01 2021
Jun 01 2022
Dec 01 2022
Jun 01 2023
Dec 01 2023
Jun 03 2024
Nov 04 2024

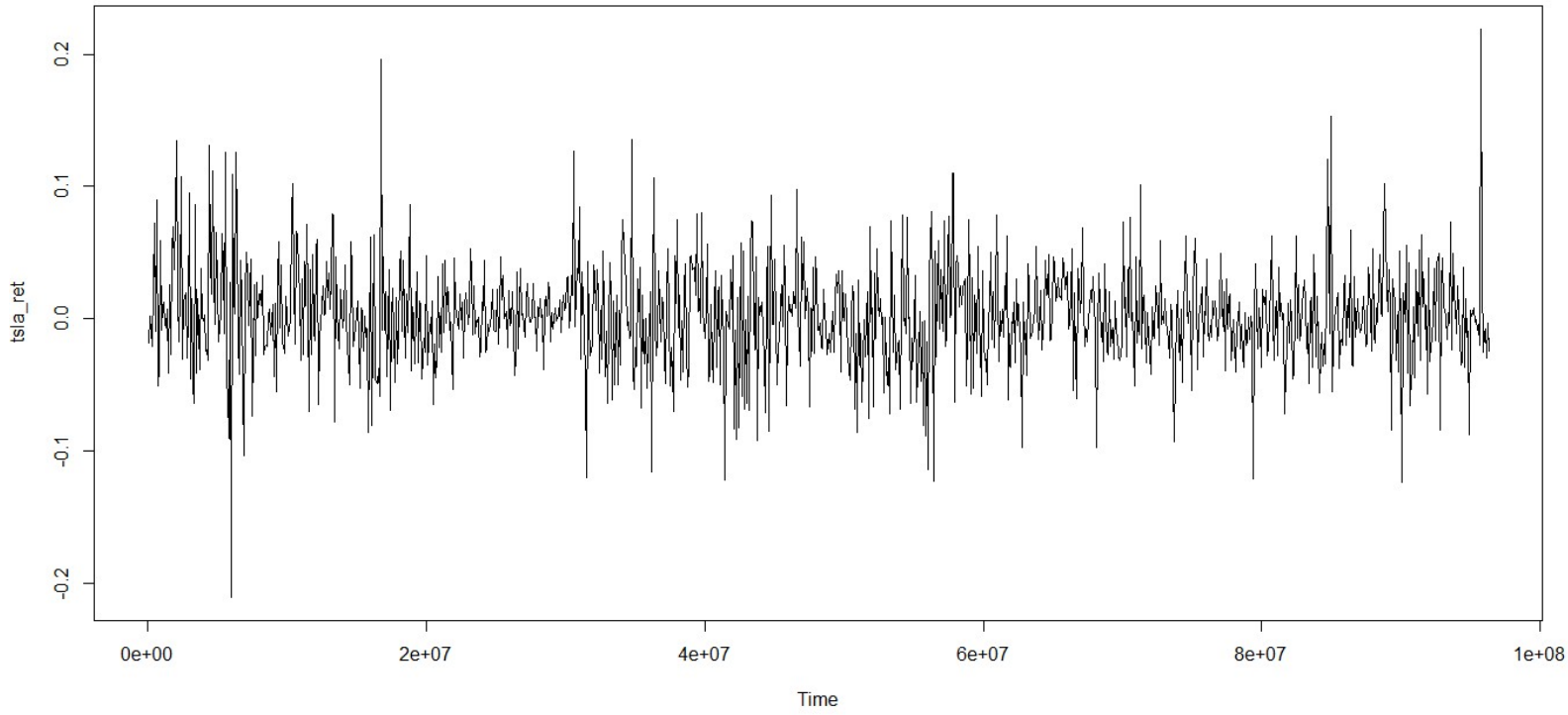
# TESLA_df_tesla$Close

```
chartSeries(df_tesla$TSLA.Close,type="bars",name="TESLA Hisse Fiyati",theme = 'white',subset='2020::2024')
```

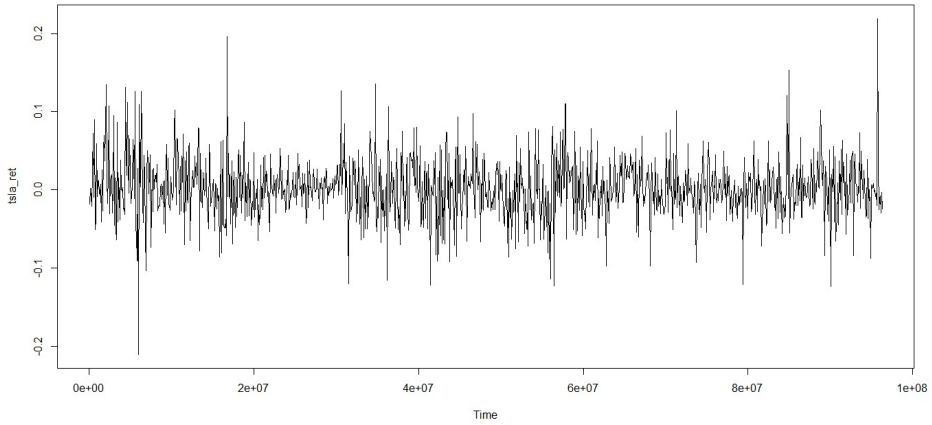


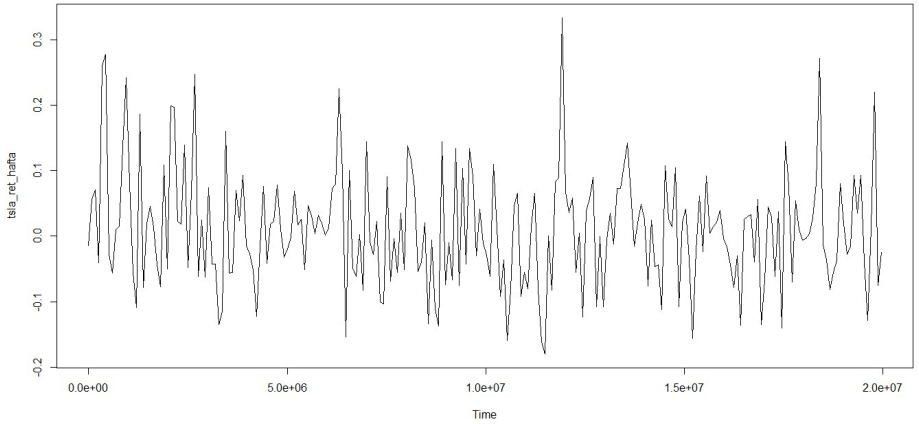
# TESLA_basit getiri (günlük)

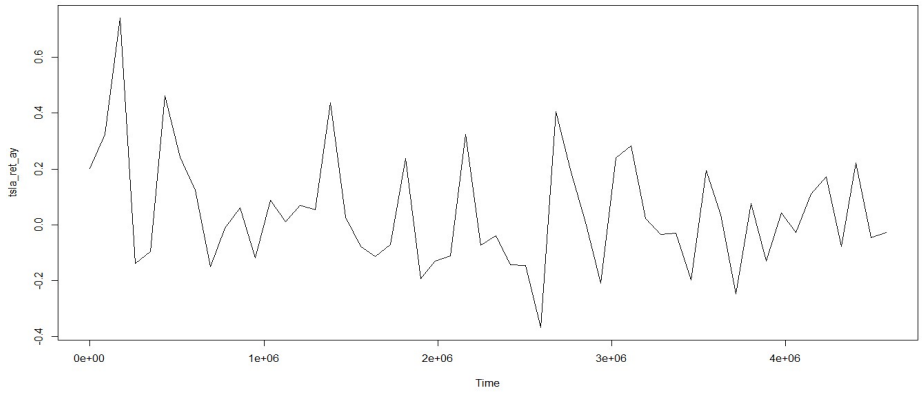
- `> tsla_ret <- CalculateReturns(df_tesla$TSLA.Close, method = "simple")`
- `> plot.ts(tsla_ret)`

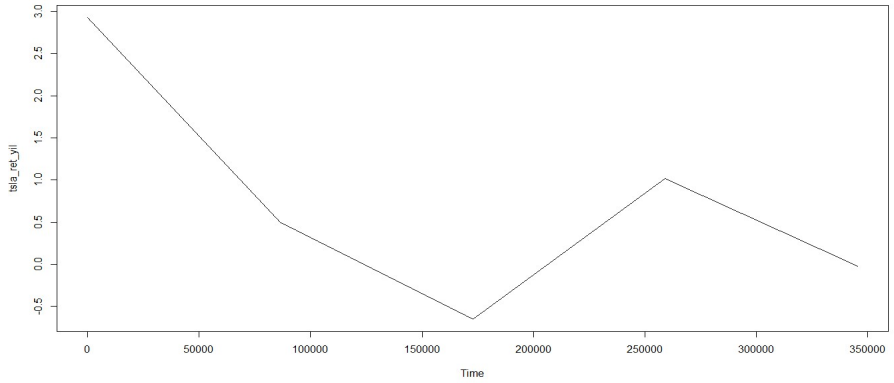


# TESLA_basit getiri



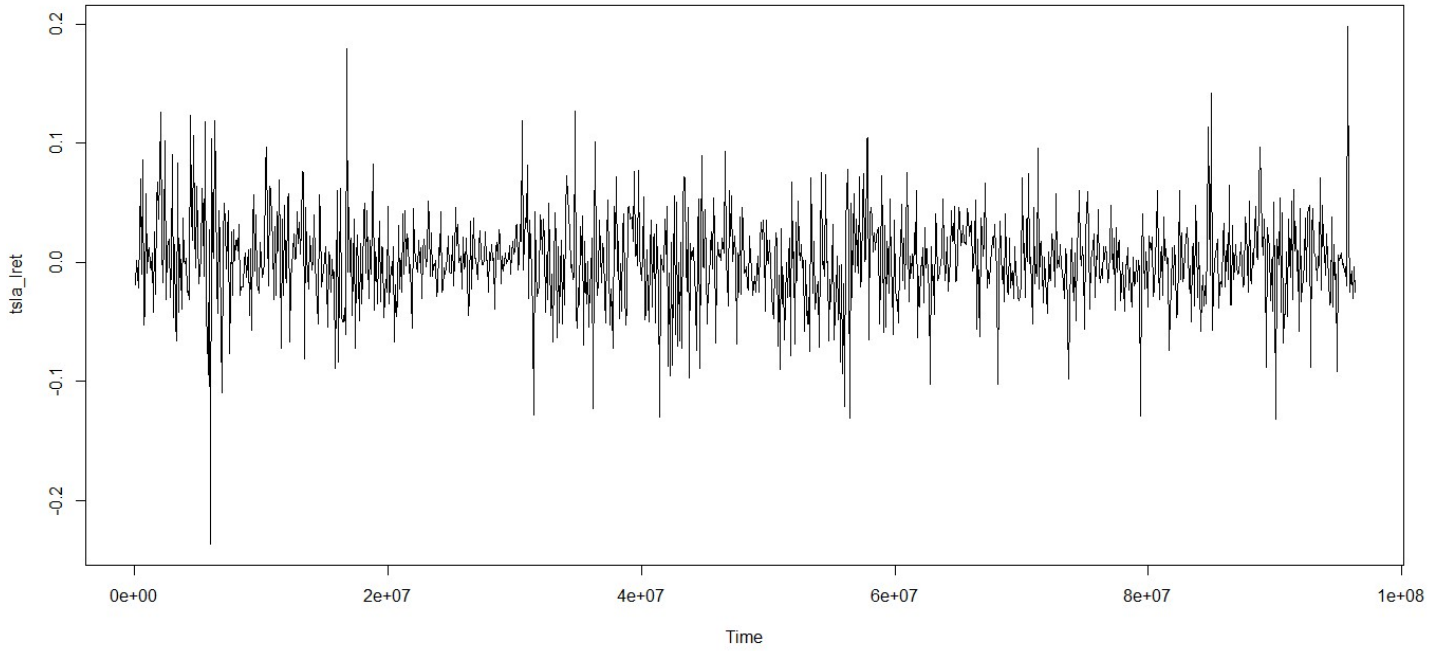



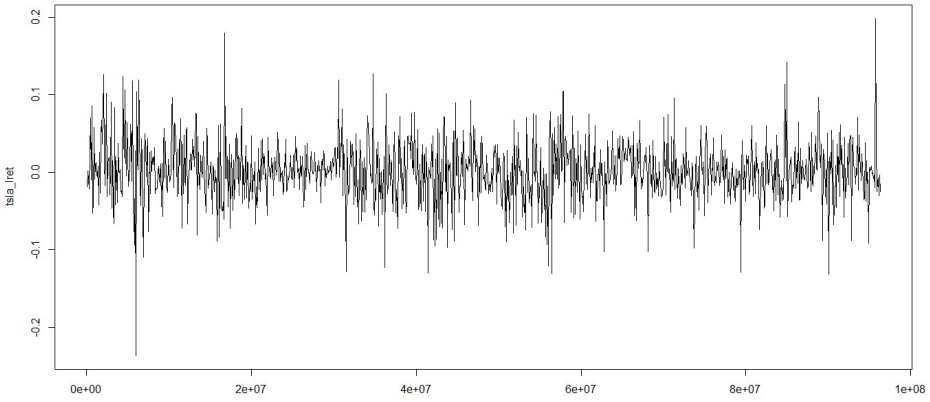



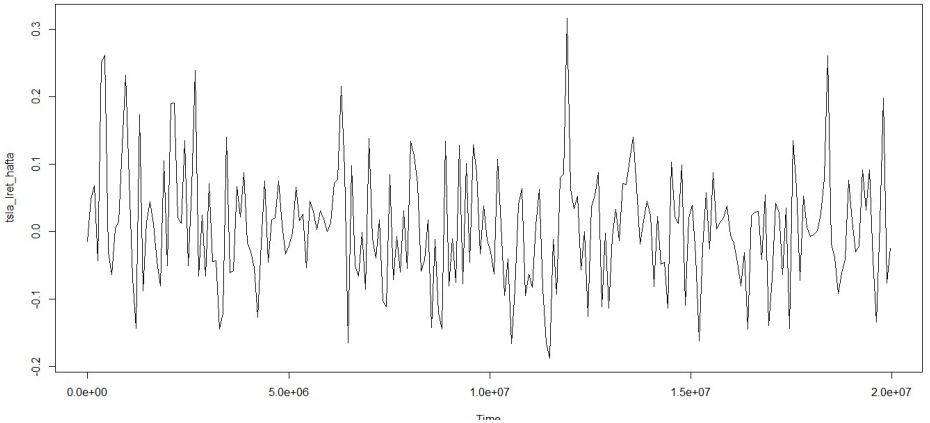
# TESLA_log_getiri

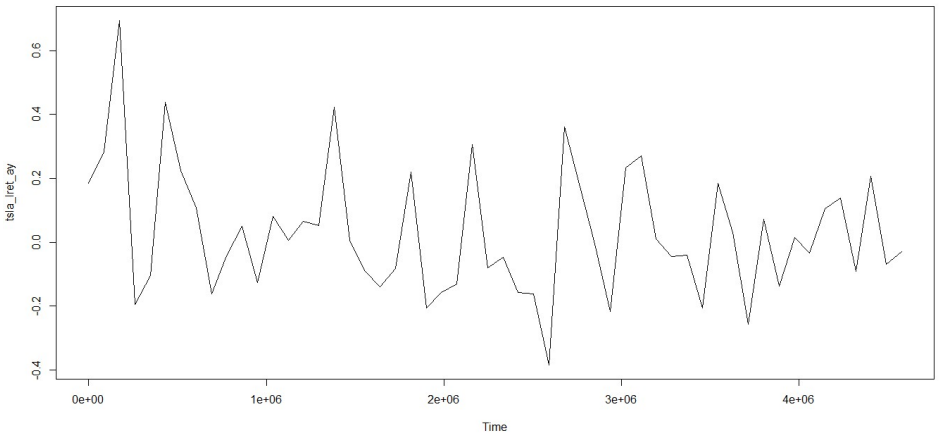
```
> tsla_lret <- CalculateReturns(df_tesla$TSLA.Close, method = "log")
> plot.ts(tsla_lret)
```

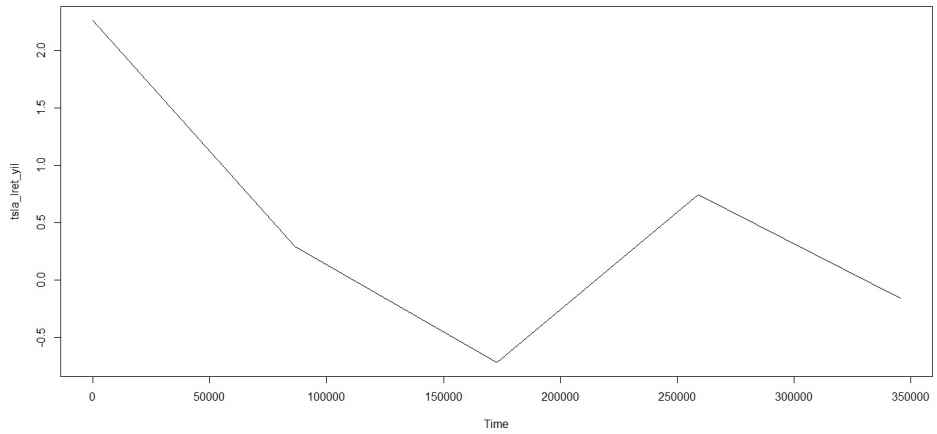


# TESLA_log_getiri



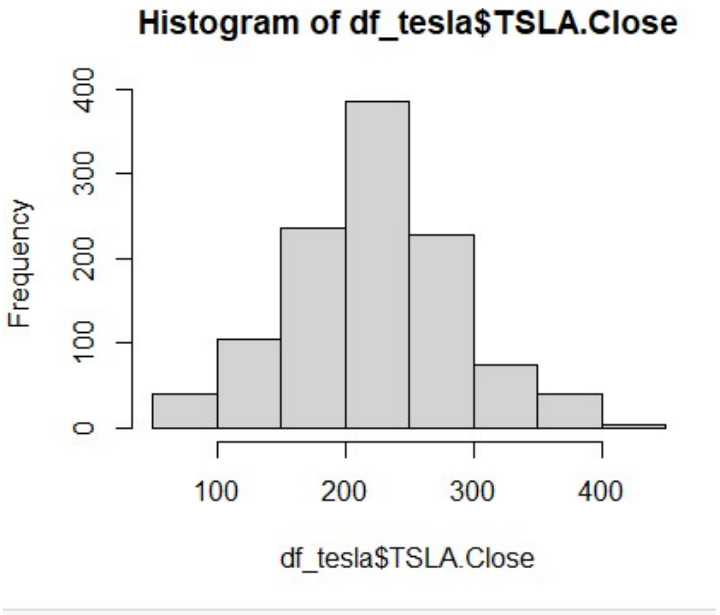



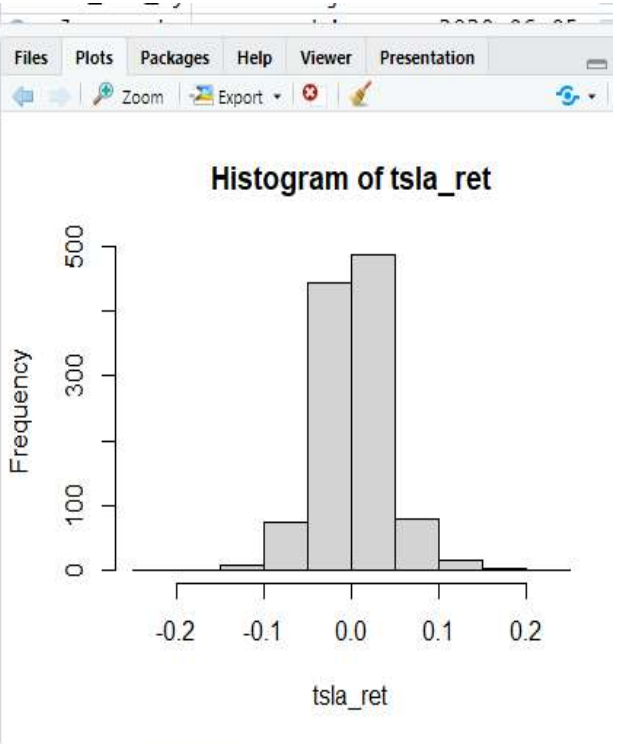
# TESLA_df_tesla$Close

```
> FinTS.stats(df_tesla$TSLA.Close)
       Start Size     Mean Standard.Deviation   Skewness Excess.Kurtosis
1 2020-06-01 1116 223.4079           63.96262 0.03903541       0.2740286
   Minimum Maximum
1 57.62533  409.97
> basicStats(df_tesla$TSLA.Close)
               TSLA.Close
nobs         1.116000e+03
NAs          0.000000e+00
Minimum      5.762533e+01
Maximum      4.099700e+02
1. Quartile  1.832575e+02
3. Quartile  2.594750e+02
Mean         2.234079e+02
Median       2.236833e+02
Sum          2.493232e+05
SE Mean      1.914671e+00
LCL Mean     2.196511e+02
UCL Mean     2.271646e+02
Variance     4.091217e+03
Stdev        6.396262e+01
Skewness     3.903500e-02
Kurtosis     2.740290e-01
>
```



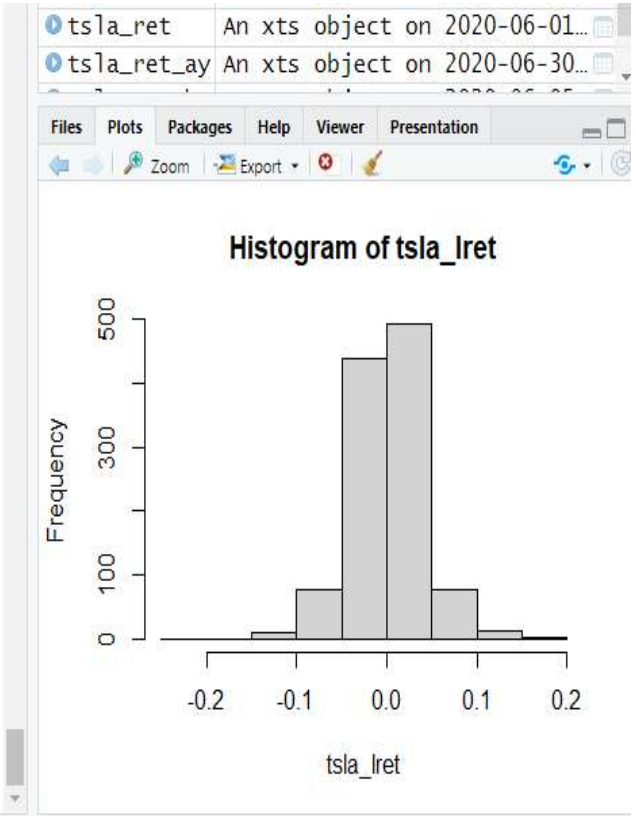
# TESLA_getiri

```
> FinTS.stats(100*tsla_ret)
       Start Size      Mean Standard.Deviation  Skewness Excess.Kurtosis   Minimum
1 2020-06-01 1115 0.2020788           3.920164 0.2490651        2.799489 -21.06282
   Maximum
1 21.91904
> basicStats(tsla_ret, ci = 0.95)
             TSLA.Close
nobs        1116.000000
NAs            1.000000
Minimum       -0.210628
Maximum        0.219190
1. Quartile   -0.020084
3. Quartile    0.021079
Mean           0.002021
Median         0.001659
Sum            2.253178
SE Mean        0.001174
LCL Mean      -0.000283
UCL Mean       0.004324
Variance       0.001537
Stdev          0.039202
Skewness       0.249065
Kurtosis       2.799489
```



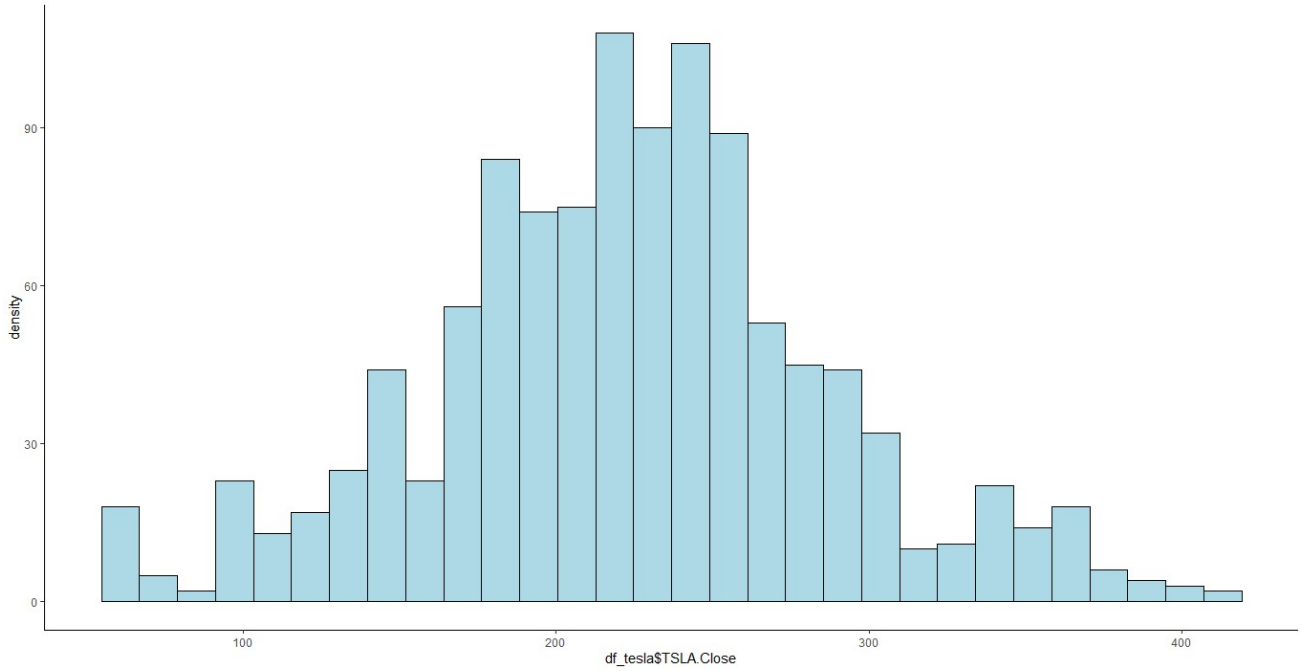
# TESLA_log_getiri

```
> FinTS.stats(100*tsla_lret)
      Start Size      Mean Standard.Deviation    Skewness Excess.Kurtosis   Minimum
1 2020-06-01 1115 0.1255759            3.90819 -0.02968084        2.813887 -23.65179
  Maximum
1 19.8187
> basicStats(tsla_lret, ci = 0.95)
             TSLA.Close
nobs       1116.000000
NAs           1.000000
Minimum      -0.236518
Maximum       0.198187
1. Quartile  -0.020288
3. Quartile   0.020860
Mean          0.001256
Median        0.001658
Sum           1.400172
SE Mean       0.001170
LCL Mean     -0.001041
UCL Mean      0.003552
Variance      0.001527
Stdev         0.039082
Skewness     -0.029681
Kurtosis      2.813887
> hist(tsla_lret)
>
```



# TESLA_getiri

```
> ggplot(data.frame(df_tesla$TSLA.Close), aes(x=df_tesla$TSLA.Close)) + +
geom_histogram(bins=30, fill="lightblue", col="black") + + geom_density(fill="red",
alpha=0.3) + + theme_classic()
```



# TESLA_df_tesla$Close

```
> #Orjinal veri icin
> jarqueberaTest(df_tesla$TSLA.Close)

Title:
 Jarque - Bera Normalality Test

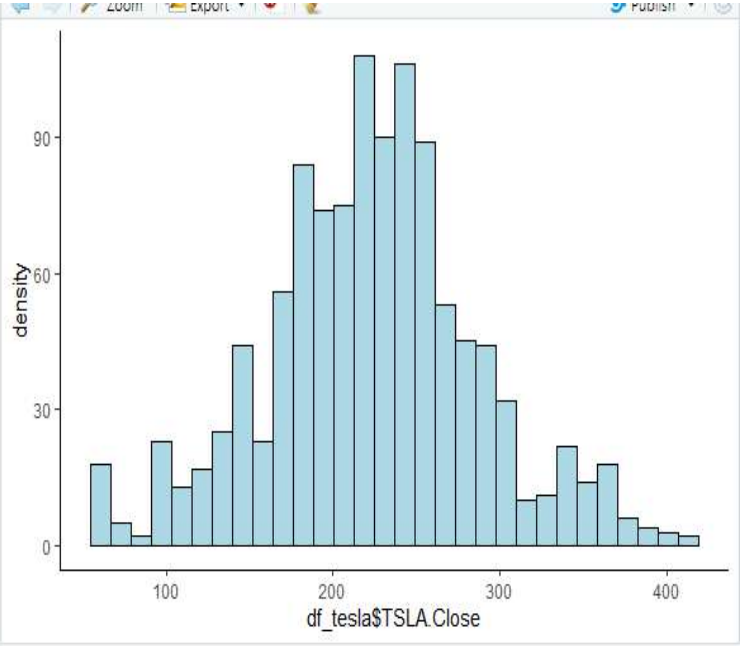
Test Results:
  STATISTIC:
    X-squared: 3.9273
  P VALUE:
    Asymptotic p Value: 0.1403

> shapiro.test(ts(df_tesla$TSLA.Close))

        Shapiro-Wilk normality test

data:  ts(df_tesla$TSLA.Close)
W = 0.99141, p-value = 4.236e-06

>
```



# TESLA_df_tesla$Close Ortalama Eşit mi?

H0: true mean is equal to 0

H0: true location shift is equal to 0

```
> t<-matrix(1:1116,1116,1)
> grup<-ifelse(t<=(nrow(t)/2),0,1)
> t.test(df_tesla$TSLA.Close~grup)

        Welch Two Sample t-test

data:  df_tesla$TSLA.Close by grup
t = 5.2897, df = 840.22, p-value = 1.564e-07
alternative hypothesis: true difference in means between group 0 and group 1 is not equal to 0
95 percent confidence interval:
 12.58841 27.44217
sample estimates:
mean in group 0 mean in group 1
       233.4155        213.4002
```

# TESLA_getiri

```
> ggplot(data.frame(tsla_ret), aes(x=tsla_ret)) + + geom_histogram(bins=30, fill="lightblue", col="black") + + geom_density(fill="red", alpha=0.3) + + theme_classic()
```

Console Terminal Background Jobs

R 4.3.1 · H:/FinansalZamanSerisi/24-25_fzsa/

```
Title:
 Jarque - Bera Normalality Test

Test Results:
  STATISTIC:
    X-squared: 378.374
  P VALUE:
    Asymptotic p Value: < 2.2e-16

> shapiro.test(ts(tsla_ret[2:1116]))

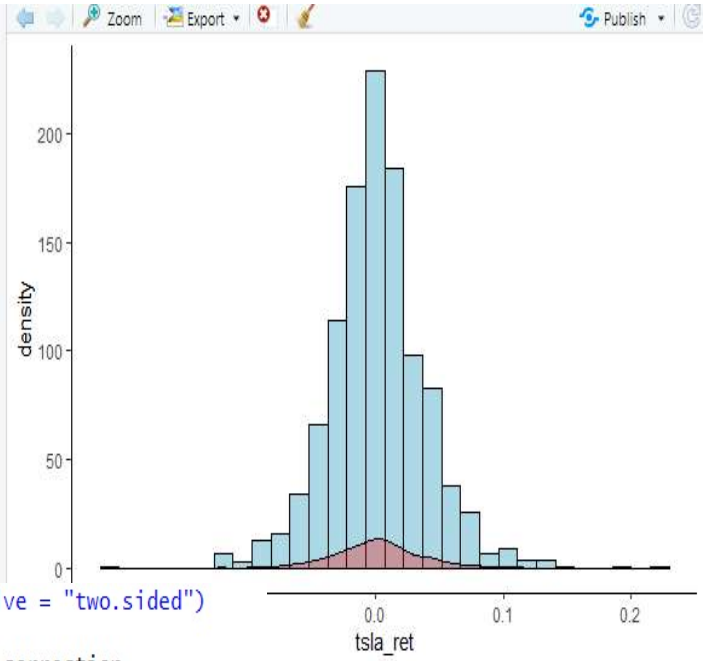
        Shapiro-Wilk normality test

data:  ts(tsla_ret[2:1116])
W = 0.9708, p-value = 3.376e-14

> wilcox.test(x = tsla_ret1, median = 0, alternative = "two.sided")

        Wilcoxon signed rank test with continuity correction

data:  tsla_ret1
V = 325776, p-value = 0.1557
alternative hypothesis: true location is not equal to 0
>
```



# TESLA_loggetiri

```
> skew.test

        D'Agostino skewness test

data:  df_tesla$TSLA.Close
skew = 1.1636, z = 23.1604, p-value < 2.2e-16
alternative hypothesis: data have a skewness
```

```
> skew.test_ret

        D'Agostino skewness test

data:  tsla_ret
skew = 0.36491, z = 8.69872, p-value < 2.2e-16
alternative hypothesis: data have a skewness
```

```
> skew.test_lret

        D'Agostino skewness test

data:  tsla_lret
skew = -0.002791, z = -0.068620, p-value = 0.9453
alternative hypothesis: data have a skewness
```

# df_tesla$TSLA.Close NORMAL???

```
> gofTest(df_tesla$TSLA.Close)    #Normal dagilim icin

Results of Goodness-of-Fit Test
-------------------------------

Test Method:                     Shapiro-Wilk GOF

Hypothesized Distribution:       Normal

Estimated Parameter(s):          mean =  78.36137
                                 sd   = 103.62308

Estimation Method:               mvue

Data:                            df_tesla$TSLA.Close

Sample Size:                     3613

Test Statistic:                  W = 0.7172918

Test Statistic Parameter:        n = 3613

P-value:                         0

Alternative Hypothesis:          True cdf does not equal the
                                 Normal Distribution.
```

# df_tesla$TSLA.Close logNORMAL???

```
> gofTest(df_tesla$TSLA.Close,distribution = "lnorm")    #Lognormal dagilim icin

Results of Goodness-of-Fit Test
-------------------------------

Test Method:                     Shapiro-Wilk GOF

Hypothesized Distribution:       Lognormal

Estimated Parameter(s):          meanlog = 3.177624
                                 sdlog   = 1.706948

Estimation Method:               mvue

Data:                            df_tesla$TSLA.Close

Sample Size:                     3613

Test Statistic:                  W = 0.9063318

Test Statistic Parameter:        n = 3613

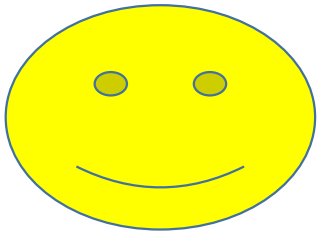
P-value:                         0

Alternative Hypothesis:          True cdf does not equal the
                                 Lognormal Distribution.
```

# TESLA_df_tesla$Close Varyanslar Eşit mi?

H0: Varyanslar eşittir

```
> library(car)
> veri<-ts(df_tesla$TSLA.Close$TSLA.Close)
> orta_deger <- nrow(veri) %/% 2
> ornek1 <- veri[1:orta_deger]  # Birinci Orneklem
> ornek2 <- veri[(orta_deger + 1):nrow(veri)]  # Birinci Orneklem
> gruplar <- factor(rep(0:1, c(length(ornek1), length(ornek2))))
> levene_testi <- leveneTest(c(ornek1, ornek2), gruplar)
> print(levene_testi)
Levene's Test for Homogeneity of Variance (center = median)
        Df F value    Pr(>F)
group    1  5378.7 < 2.2e-16 ***
      3379
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```



# TESLA_tsla_ret Varyanslar Eşit mi?

H0: Varyanslar eşittir

```
> veri_ret<-ts(tsla_ret[2:3381])
> orta_deger <- length(veri_ret) %/% 2
> ornek1 <- veri_ret[1:orta_deger]  # Birinci Orneklem
> ornek2 <- veri_ret[(orta_deger + 1):length(veri_ret)]  # Birinci O
neklem
> gruplar <- factor(rep(0:1, c(length(ornek1), length(ornek2))))
> levene_testi <- leveneTest(c(ornek1, ornek2), gruplar)
> print(levene_testi)
Levene's Test for Homogeneity of Variance (center = median)
        Df F value    Pr(>F)
group    1  28.113 1.218e-07 ***
      3376
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

# TESLA_tsla_lret Varyanslar Eşit mi?

H0: Varyanslar eşittir

```
> veri_lret<-ts(tsla_lret[2:3381])
> orta_deger <- length(veri_lret) %/% 2
> ornek1 <- veri_lret[1:orta_deger]  # Birinci Orneklem
> ornek2 <- veri_lret[(orta_deger + 1):length(veri_lret)]  # Birinci
Orneklem
> gruplar <- factor(rep(0:1, c(length(ornek1), length(ornek2))))
> levene_testi <- leveneTest(c(ornek1, ornek2), gruplar)
> print(levene_testi)
Levene's Test for Homogeneity of Variance (center = median)
        Df F value    Pr(>F)
group    1  27.711 1.497e-07 ***
      3377
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
>
```

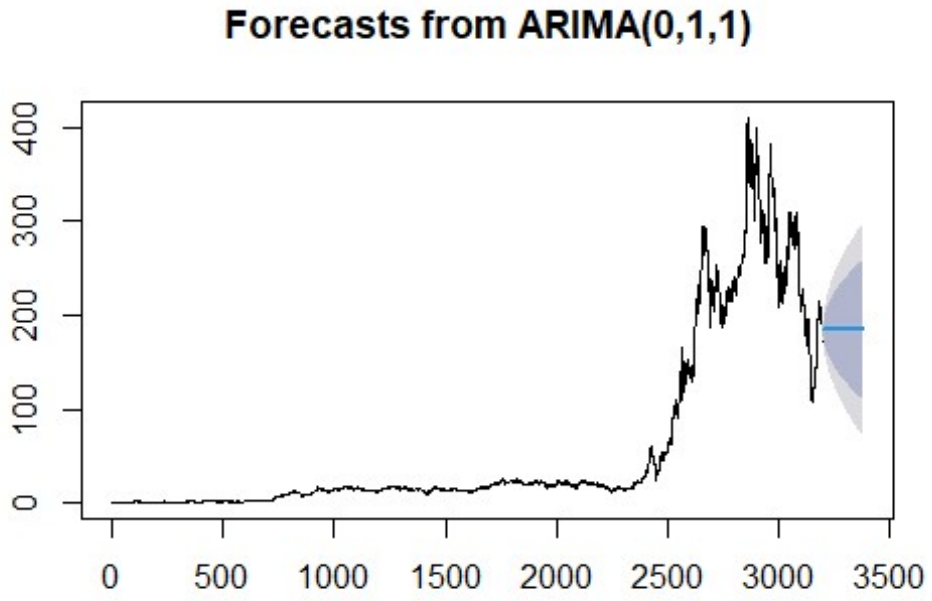
# TESLA_df_tesla$Close

```
> split_seri1 <- ts_split(ts.tsla, sample.out = 180) #son 180 değer dışarıda bırak
ılacak.
> training <- split_seri1$train
> testing <- split_seri1$test
> length(ts.tsla)
[1] 3381
> length(training)
[1] 3201
> length(testing)
[1] 180
> fit_tsla<-auto.arima(training)
> summary(fit_tsla)
Series: training
ARIMA(0,1,1)

Coefficients:
          ma1
      -0.0381
s.e.   0.0174

sigma^2 = 19.98:  log likelihood = -9331.91
AIC=18667.82   AICc=18667.82   BIC=18679.96

Training set error measures:
                     ME     RMSE      MAE        MPE     MAPE      M.
Training set 0.05924086 4.468828 1.735627 0.08674474 2.485805 0.9983426
                     ACF1
Training set -0.000760942
```



# TESLA_df_tesla$Close

```
> fit_ets<-ets(training)
> summary(fit_ets)
ETS(M,A,N)

Call:
 ets(y = training)

  Smoothing parameters:
    alpha = 0.9844
    beta  = 0.0027

  Initial states:
    l = 1.5867
    b = 0.0058

  sigma:  0.0362

     AIC     AICc      BIC
23125.81 23125.83 23156.16

Training set error measures:
                      ME     RMSE      MAE          MPE     MAPE      MASE
Training set -0.00699365 4.473396 1.734272 -0.01822253 2.486611 0.9975634
                    ACF1
Training set -0.02464476
> plot(forecast(fit_ets,h=180))
```

